TABLE 1
CHARACTERISTICS OF COMMONLY USED RECHARGEABLE BATTERIES

Specifications	Lead Acid	Ni-Cd	Ni-MH	Li-ion		
				Cobalt	Manganese	Phosphate
Specific energy (Wh/kg)	30-50	45-80	60-120	150-250	100-150	90-120
Internal resistance	Very low	Very low	Low	Moderate	Low	Very low
Cycle life (80% DoD)	200-300	1,000	300-500	500-1,000	500-1,000	1,000-2,000
Charge time	8-16h	1-2h	2-4h	2-4h	1-2h	1-2h
Overcharge tolerance	High	Moderate	Low	Low. No trickle charge		
Self-discharge/month (room temperature)	5%	20%	30%	<5% Protection circuit consumes 3%/month		
Cell voltage (nominal)	2V	1.2V	1.2V	3.6V	3.7V	3.2-3.3V
Charge cutoff voltage (V/cell)	2.40 float 2.25	Full charge detection by voltage signature		4.20 typical; Some go to higher V		3.60
Discharge cutoff voltage (V/cell, 1C)	1.75V	1.00V		2.50-3.00V		2.50V
Peak load current Best result	5C 0.2C	20C 1C	5C 0.5C	2C <1C	>30C <10C	>30C <10C
Charge temperature	-20 to 50°C (-4 to 122°F)	0 to 45°C (32 to 113°F)		0 to 45°C (32 to 113°F)		
Discharge temperature	-20 to 50°C (-4 to 122°F)	-20 to 65°C (-4 to 149°F)		-20 to 60°C (-4 to 140°F)		
Maintenance requirement	3-6 months (topping chg.)	Full discharge every 90 days when in full use		Maintenance-free		
Safety requirements	Thermally stable	Thermally stable, fuse protection		Protection circuit mandatory		
In use since	Late 1800s	1950	1990	1991	1996	1999
Toxicity	Very high	Very high	Low	Low		
Coulombic efficiency	~90%	~70% slow charge ~90% fast charge		99%		
Cost	Low	Moderate		High		

(Source: batteryuniversity.com)

scooter, or motorcycle. Each of them needs a battery with different chemistry and a capacity of 12V, 24V, or 36V depending on the range and speed required.

The first factor to consider, of course, is cost, which may also decide the quality. The other important metric is specific energy density that tells the amount of energy contained per kg unit of the battery. For instance, a lightweight vehicle requires a battery with high specific energy density of about 1,000Wh/kg.

Volume energy density is the metric that needs attention for fixing a battery in a small space. While retrofitting a 4-wheeler or some other vehicle, there might be space constraint to replace the engine and associated components with motor controller and power electron. Here opting for battery cells with high volume energy density helps. For example, if one litre (or 0.01m^3) volume can hold more energy, it means that much lesser space is needed.

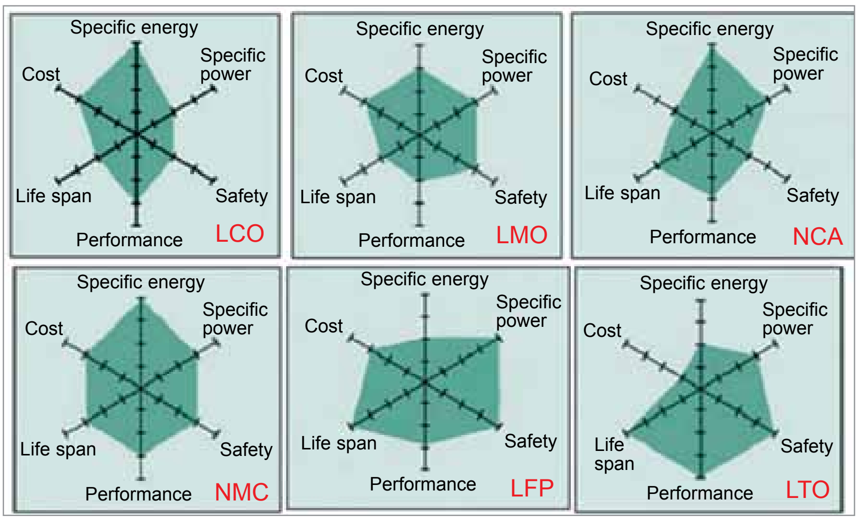

Fig. 1: Spider diagram for six most relevant Li-ion battery technologies (Source: Boltta Battery Technologies)